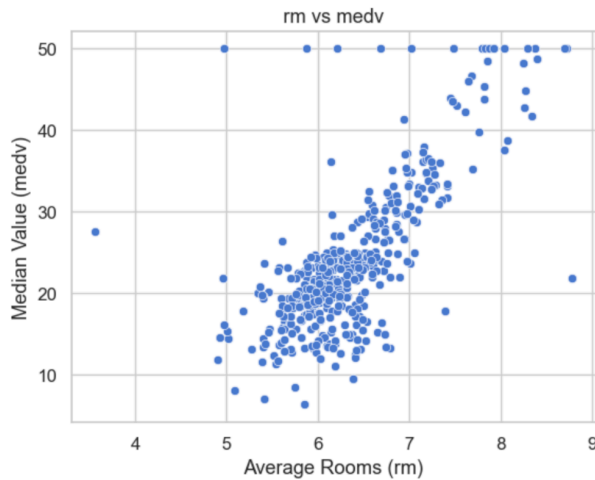


Assignment - 6

2. Perform bivariate visualizations between the median value of owner-occupied homes, medv and the following variables on the dataset: rm, age, dis, rad and ptratio. Please use different Jupyter Notebook cell for each bivariate visualization, and briefly comment on your results

RM vs MEDV

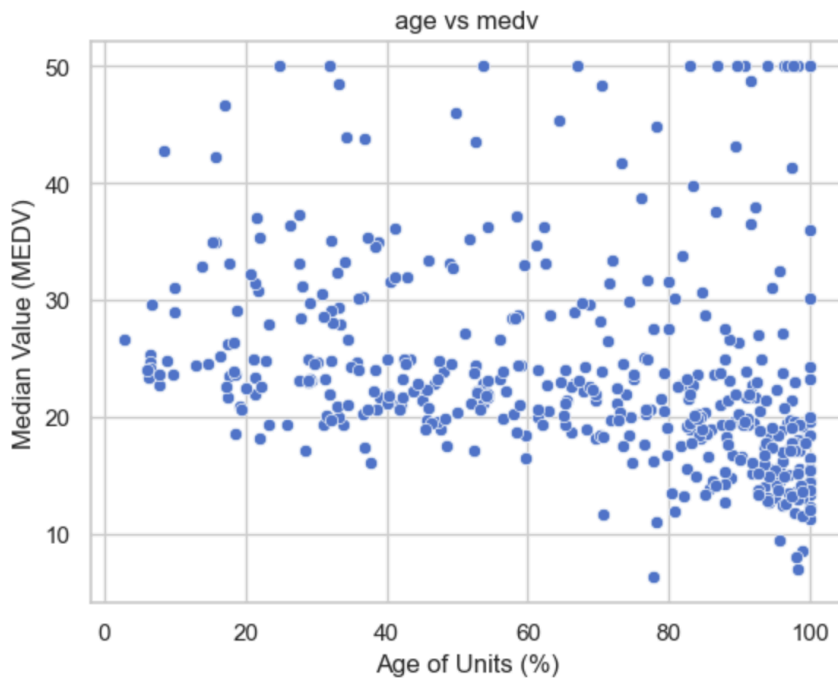
```
[38]: # Displaying scatterplot for rm vs medv
sns.scatterplot(x=df['rm'], y=df['medv'])
plt.title("rm vs medv")
plt.xlabel("Average Rooms (rm)")
plt.ylabel("Median Value (medv)")
#Display plot
plt.show()
```



We can see a significant increasing pattern in home with more rooms being closely associated with high median values. This says that larger homes seem to be more attractive that indicate rich neighborhoods.

AGE vs MEDV

```
# Displaying scatterplot for age vs medv
sns.scatterplot(x=df['age'], y=df['medv'])
plt.title("age vs medv")
plt.xlabel("Age of Units (%)")
plt.ylabel("Median Value (medv)")
#Display plot
plt.show()
```



The scatterplot shows that old houses are likely to have lower values, however the relation isn't precisely linear. This could be because of decreases in value, old infrastructure, and outdated interior designs that diminishing demand in the market.

DIS vs MEDV

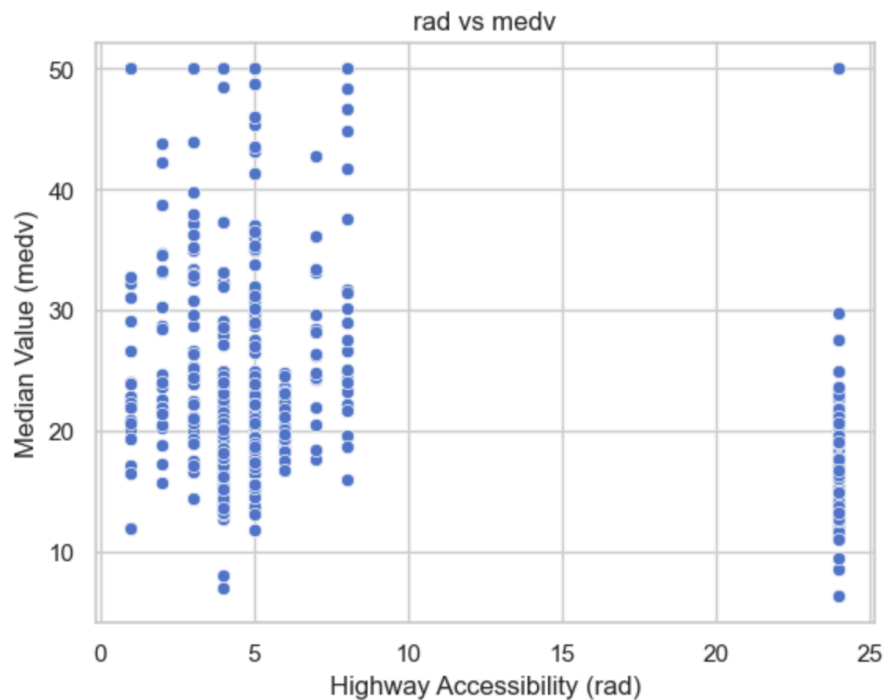
```
# Displaying scatterplot for dis vs medv
sns.scatterplot(x=df['dis'], y=df['medv'])
plt.title("dis vs medv")
plt.xlabel("Distance to Employment Centers (dis)")
plt.ylabel("Median Value (medv)")
# Display plot
plt.show()
```



Homes which are located farther away from employment centers (higher DIS) tend to have higher median values. It additionally suggests that rural areas or less crowded locations could be wealthier, since families opt for space and commute into urban cities.

RAD vs MEDV

```
# Displaying scatterplot for rad vs medv
sns.scatterplot(x=df['rad'], y=df['medv'])
plt.title("rad vs medv")
plt.xlabel("Highway Accessibility (rad)")
plt.ylabel("Median Value (medv)")
#Display plot
plt.show()
```

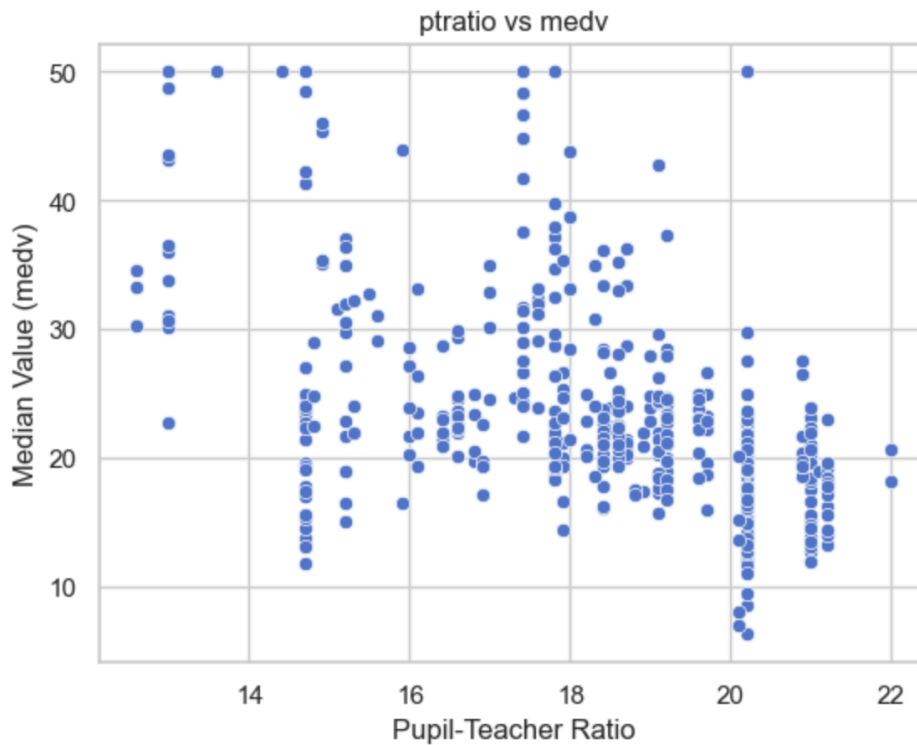


The median home values vary drastically among various levels of highway accessibility, yet specific categories such as RAD=24 have lower values.

This could suggest that though accessibility is essential, living close enough to a highway might reduce desirability due to various pollutions.

PTRATIO vs MEDV

```
# Displaying scatterplot for ptratio vs medv
sns.scatterplot(x=df['ptratio'], y=df['medv'])
plt.title("ptratio vs medv")
plt.xlabel("Pupil-Teacher Ratio")
plt.ylabel("Median Value (medv)")
#Display plot
plt.show()
```



The scatterplot depicts a negative trend which shows higher pupil-teacher ratios (crowded schools) have a correlation with lower home values. This means that costs of housing are greatly impacted by school quality, with families spending higher to stay in neighborhoods with quality educational systems.